Embeddings from TPCx-IoT data: 10 substations to one node with NebulaStream

Demo report · LLMs in Hierarchical Aggregation Topologies · Vasil · 12 November 2026

1. What was run

The TPCx-IoT kit (v2.1.2) models an electric utility whose power substations send sensor readings to IoT gateways. Its data generator (a modified YCSB) was run with 10 clients, one per substation, and its 207,989 records were stored in SQLite. Each substation is one NebulaStream edge worker (A1–A10). Every 60 s window of a substation becomes one embedding: the mean reading of each of its 160 sensors in that window, standardised and reduced with one shared PCA to 32 dimensions (unit length). Each worker sends its 35 windows to a single node B, which computes `SCORE` = cosine similarity to the centroid of all windows in SQL inside NebulaStream (the edge workers have `max_operators: 0`, so the computation is placed on B).

2. What the generator produces

A record is a key and one 1,000-character value, for example:

```
client1:ang_30_vibration:1794503471655
ang_30_vibration:ang_30_vibration_value:0.3218:timestamp:1794503481566:455679879:ang_30_vi…
```

Each record is one randomly chosen sensor (160 sensors: 13 locations × 13 measurement types such as Power, Temperature and Pressure), and timestamps advance 100 ms per record (about 2,080 s per substation). The value contains the reading, a 4-decimal number, and repeats it as filler up to 1,000 characters (about 1,038 bytes per record in total). The readings themselves have no structure:

Distribution	mean 0.499, std 0.289; indistinguishable from uniform [0, 1) (KS test p = 0.18)
Memory over time	lag-1 autocorrelation per sensor -0.008 (0 = independent readings)
Substations differ?	no (ANOVA across the 10 clients, p = 0.27)
Measurement types differ?	no: Power looks like Temperature (ANOVA, p = 0.16)

3. Did everything arrive?

Windows received at B	350 of 350 (every worker 35/35)
Embeddings at B vs. as sent	bit-identical (max. difference 0)
SCORE at B vs. Python	max. difference 5.2e-07 (float32 rounding)
Bytes sent as embeddings	49 KB (144 B per window)
Same windows as generated records	210,863 KB → 4,284× more (mostly filler)
Same windows, key + reading only	8,967 KB → 182× more (the fair comparison)

Transport works exactly as with the 20 Newsgroups demo. The saving is large because one embedding replaces a whole window of about 600 records; against the generated records it is inflated by the generator's filler, so the content-only figure (182×) is the one to quote.

4. What B received, coloured by its score



Figure 1. Each point is one of the 350 windows B received, projected from 32 to two dimensions with PCA (no units). Colour is the SCORE computed by B.

Unlike the 20 Newsgroups demo, there is no region of high scores: the points form one round cloud and the colours are mixed throughout. B's mean score per substation lies between -0.024 and +0.036, i.e. no substation stands out. The PCA reduction confirms there is nothing to find: the 32 dimensions keep 41.4% of the variance, the same as for pure noise of the same shape (41.5%).

5. Each substation's windows

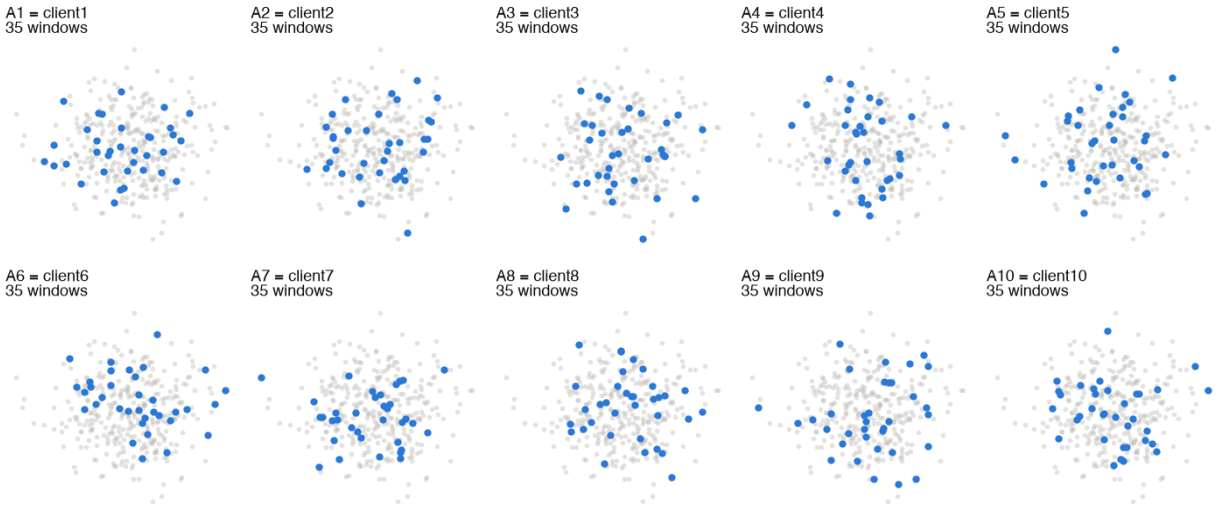


Figure 2. The same projection, drawn once per edge worker: its windows in blue, all others in grey.

Every substation's windows are spread over the whole cloud, overlapping all others. Measured: a nearest-neighbour classifier recognises the sending substation from its embedding 11% of the time, against 10% by chance (for comparison, the 20 Newsgroups embeddings identified the newsgroup 71% of the time). With random readings this is the expected result: the embeddings are correct, there is simply no structure in the data for them to preserve.

6. Conclusion and proposal

With TPCx-IoT data the pipeline delivers all 350 window embeddings from 10 substations to one node, unchanged, with B computing on them inside NebulaStream, and it needs 182× fewer bytes than the

windows' actual content. Whether embeddings preserve meaning cannot be tested on this data, because the generator's readings are uniform random numbers with no differences between substations, sensors or time.

Proposal. Keep TPCx-IoT's structure (substations, sensor names, keys, rates; this maps directly onto edge workers $\rightarrow$ gateways $\rightarrow$ sink, e.g. 10 $\rightarrow$ 3 $\rightarrow$ 1) but give each substation realistic behaviour (its own levels, daily cycles, correlated sensors) and inject labelled anomalies, so that the embeddings and B's anomaly score can be evaluated against ground truth. The kit's licence allows modifying the generator. Alternatively, use real smart-grid data with the same shape (e.g. DEBS 2014).